

[K-Means and K-Medoids clustering]

#	point	Height - cm	Weight - Kg	Shoe Size
	P1	170	65	42
	P2	180	80	44
	P3	160	55	40
	P4	165	60	41
	P5	155	50	39
	P6	172	70	43
	P7	158	52	39
	P8	185	85	45

$K=2$

1) K-Means clustering

$$c_1^{(0)} = P1 = (170, 65, 42)$$

$$c_2^{(0)} = P2 = (180, 80, 44)$$

$$d(P_i, c_j) =$$

$$\sqrt{(h_i - h_j)^2 + (w_i - w_j)^2 + (s_i - s_j)^2}$$

point	$d(P_i, c_1^{(0)})$	$d(P_i, c_2^{(0)})$
P1	0	18.138
P2	18.138	0
P3	14.283	32.269
P4	7.141	25.180
P5	21.424	39.370
P6	5.477	12.845
P7	17.944	35.966
P8	25.180	7.141

cluster 1:

{P1, P3, P4, P5,
P6, P7 }

cluster 2: {P2, P8 }

Cluster 1 mean:

$$\text{Height mean} = \frac{170 + 160 + 165 + 172 + 158}{6} = 163.33$$

$$\text{weight mean} = \frac{65 + 55 + 60 + 50 + 70 + 52}{6} = 58.67$$

$$\text{shoe mean} = \frac{42 + 40 + 41 + 39 + 43 + 39}{6} = 40.67$$

$$c_1^{(1)} = (163.33, 58.67, 40.67)$$

Cluster 2 mean:

$$\text{Height mean} = \frac{180 + 185}{2} = 182.5$$

$$\text{weight mean} = \frac{80 + 85}{2} = 82.5$$

$$\text{shoe mean} = \frac{44 + 45}{2} = 44.5$$

$$c_2^{(1)} = (182.5, 82.5, 44.5)$$

2] k-medoids clustering (PAM Algorithm)

$$m_1^{(0)} = p_1 = (170, 65, 42)$$

$$m_2^{(0)} = p_2 = (180, 80, 44)$$

[Previously done in k-means clustering
Exactly same table]

→ Cluster A : {p1, p3, p4, p5, p6, p7}

Cluster B : {p2, p8}

$$\text{cost}^{(0)} = \sum_i d(p_i, m_{\text{assigned}})$$

$$= \cancel{18.138} + 0 + 14.283 + 7.41 + 21.424 + 5.477 + 17.944 + 0 + 7.14 + 8.58 = 73.416$$

$$\approx 73.416$$

Now,

$$m_1 = p_3$$

$$m_1^{(1)} = p_3 = (160, 55, 40)$$

$$m_2^{(1)} = p_2 = (180, 80, 44)$$

Point	$d(P_i, M_1^{(n)})$	$d(P_i, M_2^{(n)})$
P1	✓ 14.28	18.138
P2	32.26	0 ✓
P3	✓ 0	32.269
P4	✓ 7.14	25.180
P5	✓ 7.14	39.370
P6	19.44	12.845 ✓
P7	✓ 3.74	35.966
P8	39.37	7.14 ✓

Cluster A: {P1, P3, P4, P5, P7}

Cluster B: {P2, P6, P8}

$$\text{Cost} = 32.3 + 19.985 = 52.285$$

[Gradient Descent]

$$\hat{y} = b_0 + w_1 x_1 + w_2 x_2$$

$$L(b_0, w_1, w_2) = \sum_{i=1}^4 (y_i - \hat{y}_i)^2$$

$$\frac{\partial L}{\partial b_0} = -2 \sum_{i=1}^4 (y_i - \hat{y}_i)$$

$$\frac{\partial L}{\partial w_1} = -2 \sum_{i=1}^4 x_1^{(i)} (y_i - \hat{y}_i)$$

$$\frac{\partial L}{\partial w_2} = -2 \sum_{i=1}^4 x_2^{(i)} (y_i - \hat{y}_i)$$

Dataset:

i	x_1	x_2	y	$y - \hat{y}_i^{(0)}$	$(y - \hat{y}_i)^2$
1	1	2	5	5	25
2	2	0	6	6	36
3	3	1	10	10	100
4	0	2	4	4	16
					177

$$\begin{array}{l} w_1^{(0)} = 0 \\ w_2^{(0)} = 0 \\ b = 0 \end{array} \quad \left| \quad \alpha = 0.01 \right.$$

$$\therefore L = 177$$

$$\begin{aligned} \frac{\partial L}{\partial b_0} &= -2 \sum_{i=1}^4 (y_i - \hat{y}_i^{(0)}) \\ &= -2(5 + 6 + 10 + 4) = -50 \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial w_1} &= -2 \sum_{i=1}^4 x_{1i} (y_i - \hat{y}_i^{(0)}) \\ &= -2(1 \times 5 + 2 \times 6 + 3 \times 10 + 0 \times 4) \\ &= -2(5 + 12 + 30) \\ &= -94 \end{aligned}$$

$$\frac{\partial L}{\partial w_2} = -2 \sum_{i=1}^4 x_2^{(i)} (y_i - \hat{y}_i^{(0)})$$

$$= -2 (2 \times 5 + 0 \times 6 + 1 \times 10 + 2 \times 4)$$

$$= -2 (10 + 10 + 8)$$

$$= -56$$

$$\therefore \nabla L^{(0)} = \begin{bmatrix} -50 \\ -94 \\ -56 \end{bmatrix}$$

$$b_0^{(1)} = 0 - 0.01 \times (-50) = 0.50$$

$$w_1^{(1)} = 0 - 0.01 \times (-94) = 0.94$$

$$w_2^{(1)} = 0 - 0.01 \times (-56) = 0.56$$

$$\pi_1^{(1)} = y - \hat{y}_i^{(1)}$$

$$= 5 - (0.50 + 0.94 \times 1 + 0.56 \times 2) = 5 - 2.56 = 2.44$$

$$\pi_2^{(1)} = 6 - (0.50 + 0.94 \times 2 + 0.56 \times 0) = 6 - 2.38 = 3.62$$

$$\pi_3^{(1)} = 10 - (0.50 + 0.94 \times 3 + 0.56 \times 1) = 10 - 3.88 = 6.12$$

$$\pi_4^{(1)} = 4 - (0.50 + 0.94 \times 0 + 0.56 \times 2) = 4 - 1.62 = 2.38$$

$$L = (2.44)^2 + (3.62)^2 + (6.12)^2 + (2.38)^2$$

$$= 62.1768$$

Logistic Regression

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$z = b_0 + w_1 x_1 + w_2 x_2$$

$$L(b_0, w_1, w_2) = \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

$$\frac{\partial L}{\partial b_0} = \sum_{i=1}^n (\hat{y}_i - y_i)$$

$$\frac{\partial L}{\partial w_j} = \sum_{i=1}^n (\hat{y}_i - y_i) x_j^{(i)}$$

Dataset

i	x_1	x_2	y	$(y - \hat{y}_i)$
1	1	2	1	0.5
2	2	0	1	-0.5
3	3	1	0	0.5
4	0	2	0	0.5

$$b_0 = 0$$

$$w_1 = 0$$

$$w_2 = 0$$

$$\alpha = 0.1$$

$$\hat{y}_i^{(0)} = \sigma(0) = \frac{1}{1+e^0} = 0.5$$

$$l_i^{(0)} = -[y_i \log(\hat{y}_i^{(0)}) + (1-y_i) \log(1-\hat{y}_i^{(0)})]$$

$$\approx 0.69$$

$$L^{(0)} = 4 \times 0.69$$

$$= 2.77$$

$$\frac{\partial L}{\partial b_0} = \sum_{i=1}^4 (\hat{y}_i^{(0)} - y_i)$$

$$= (-0.5) + (-0.5) + 0.5 + 0.5 = 0$$

$$\frac{\partial L}{\partial w_1} = \sum_{i=1}^4 x_1^{(i)} (\hat{y}_i^{(0)} - y_i)$$

$$= (-0.5) \times 1 + 2 \times (-0.5) + (0.5) \times 3 + (0.5) \times 0$$

$$= -0.5 - 1.0 + 1.5 + 0$$

$$= 0$$

$$\frac{\partial L}{\partial w_2} = 0.5$$

$$\nabla L^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0.5 \end{bmatrix}$$

$$b_0^{(1)} = 0 - 0.1 \times 0 = 0$$

$$w_1^{(1)} = 0 - 0.1 \times 0 = 0$$

$$w_2^{(1)} = 0 - 0.1 \times 0.5 = -0.05$$

$$z_1^{(1)} = b_0^{(1)} + w_1^{(1)}x_1 + w_2^{(1)}x_2$$

$$= 0 + 0 \times 1 + (-0.05) \times 2 = -0.10$$

$$\hat{y}_1^{(1)} = \sigma(-0.10) = \frac{1}{1+e^{-(-0.10)}} \approx 0.47$$

$$z_2^{(2)} = 0 + 0 \times 2 + (-0.05) \times 0 = 0$$

$$\therefore \hat{y}_2^{(2)} = \sigma(0) = 0.5$$

$$z_3^{(1)} = -0.05$$

$$\hat{y}_3^{(1)} = \sigma(-0.05) \approx 0.49$$

$$z_4^{(1)} = 0 + 0 \times 0 + (-0.05) \times 2 = -0.10$$

$$\hat{y}_4^{(1)} = \sigma(-0.10) \approx 0.48$$

$$L^{(1)} = [-\log(\hat{y}_1^{(1)}) \times 1] + [-1 \times \log(0.5)]$$

$$+ [-(1-0) \log(1 - \hat{y}_3^{(1)})] + [-(1-0) \log(1 - \hat{y}_4^{(1)})]$$

$$= -\log(0.48) - \log(0.5) - \log(1 - 0.49)$$

$$- \log(1 - 0.48)$$

$$\approx 2.75$$

(GSE437)

[Data Transformation]

Mapping $\rightarrow$

male	— 1	High School	— 1
Female	— 0	Bachelor	— 2
		Master	— 3

Encode Labels

Quantization $\rightarrow$

- 0-12 = child
- 13-19 = Teen
- 20-59 = Adult
- 60+ = Senior

Simplify ranges

(decision trees / rule-based models)

Binarization $\rightarrow$ level If income > 50 = 1 Threshold-based
Else = 0 flags

Normalization $\rightarrow$ min-max Normalization

$$\frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

[0, 1]

compare feature values (classification task)

Scaling (Standardization) $\rightarrow$

For normal distribution

1) min-max Scaling, $X_{\text{norm}} = \frac{X - \min}{X_{\max} - X_{\min}}$ [0, 1]

2) Standard Scaling (Z -score), $X_{\text{scaled}} = \frac{X - \mu}{\sigma}$

3) Robust Scaling, $X_{\text{robust}} = \frac{X - Q_2}{Q_3 - Q_1}$

outliers resistance

[Association Rule Mining]

✓ Also known as Market Basket Analysis

✓ we will use Apriori algorithm

↳ used to discover relationship between items in large dataset.

Terminologies:

- 1) Itemset: collection of items. Ex: {milk, bread}
- 2) Transaction: Set of items bought together
- 3) Association Rule: Generating rules like - If a customer buys X, he will also buy Y. The probability is P%.

Example: Dataset is given. Find Association rules.

<u>TID</u>	<u>Items purchased</u>
T1	{Bread, milk}
T2	{Bread, Diapers, Beer, Eggs}
T3	{milk, Diapers, Beer, Cola}
T4	{Bread, milk, Diapers, Beer}
T5	{Bread, milk, Diapers, Cola}

Note that, min support = 40%,
min confidence = 60%.

[Apriori Algorithm]

Solution:

① Frequent itemsets (1-itemset, C_1)

i)

Itemset	Support Count	Support (%)
{Bread}	4	$4/5 = 80\%$
{Milk}	4	$4/5 = 80\%$
{Diapers}	4	$4/5 = 80\%$
{Beer}	3	$3/5 = 60\%$
{Eggs}	1	$1/5 = 20\%$ ← Pruned
{Cola}	2	$2/5 = 40\%$

Frequent 1-itemsets (F_1): {Bread}, {Milk}, {Diapers}, {Beer}, {Cola}

ii) (2 itemset, C_2)

Itemset	Support Count	Support (%)
{Bread, Milk}	3	$3/5 = 60\%$
{Bread, Diapers}	3	60%
{Bread, Beer}	2	40%
{Bread, Eggs}		
{Bread, Cola}	1	20% ← Pruned
{Milk, Diapers}	3	60%
{Milk, Beer}	2	40%
{Milk, Eggs}		

$\{\text{milk, cola}\}$	2	40%
$\{\text{Diapers, Beer}\}$	3	60%
$\{\text{Diapers, cola}\}$	2	40%
$\{\text{Beer, cola}\}$	1	20% ← Pruned

Frequent 2-item sets (F_2) = $\{\text{Bread, milk}\},$
 $\{\text{Bread, Diapers}\},$

$\{\text{Bread, Beer}\}, \{\text{milk, Diapers}\}, \{\text{milk, Beer}\},$
 $\{\text{milk, cola}\}, \{\text{Diapers, Beer}\}, \{\text{Diapers, cola}\}$

iii) (3-itemsets (C_3))

Itemset	Support Count	Support (%)
$\{\text{Bread, milk, Diapers}\}$	2	40%
$\{\text{Bread, milk, Beer}\}$	1	20% ← Prune
$\{\text{Bread, Diapers, Beer}\}$	2	40%

{Milk, Diapers,
Beer}

2

10%

{Milk, Diapers,
cola}

1

20% ← pruned

Frequent 3-itemsets (F_3) = {Bread, milk, Diapers},
{Bread, Diapers, Beer},
{milk, Diapers, Beer}

(4-itemsets, C_4)

Itemset

Support

Support (%)

{Bread, milk,
Diapers, Beer}

1

20% ← pruned

↑
No need since at C_3 we already reached
at minimum support.

* Support: How frequently the itemset appears in the dataset.

$$= \frac{(X \cup Y). \text{count}}{n}$$

$$= \frac{\text{num of transactions containing } X \text{ and } Y}{\text{Total num of transactions}}$$

* confidence: How often the rule is true.

$$= \frac{(X \cup Y). \text{count}}{X. \text{count}}$$

$$= \frac{\text{support of } X \text{ and } Y}{\text{support of } X}$$

* Lift:

$$= \frac{\text{sup}(X \cap Y)}{\text{sup}(X) \times \text{sup}(Y)}$$

2) Generating Association Rule.

$$1) \{ \text{Bread, milk} \} \rightarrow \{ \text{Diapers} \}$$

$$\text{confidence} = \frac{\text{support}(\{ \text{Bread, milk, Diapers} \})}{\text{support}(\{ \text{Bread, milk} \})}$$

$$= \frac{2}{3}$$

$$= 66.7\% > 60\%$$



$$2) \{ \text{Bread, Diapers} \} \rightarrow \{ \text{milk} \}$$

$$\text{confidence} = \frac{\text{support}(\{ \text{Bread, Diapers, milk} \})}{\text{support}(\{ \text{Bread, Diapers} \})}$$

$$= \frac{2}{3}$$

$$= 66.7\% > 60\%$$



Interpretation

Rule: $\{ \text{Bread, milk} \} \rightarrow \{ \text{Diapers} \}$

→ customers who buy bread and milk are 66.7% likely to also buy diapers.

→ This occurs in 60% transactions.

#

~~{milk, bread}~~ $\rightarrow$ {Diapers}

$$\text{List} = \frac{\text{Support of } (\{milk, bread\})}{\text{Support of } (\{milk, bread\}) \times \text{Support of } (\{Diapers\})}$$

$$= \frac{2}{3 \times 4} = 0.16$$

association
↓
greater than
1 → Independent

> 1.5
↑
real association

KNN

(K=3)

#

ID	Age	Salary (K)	Has_wife (Label)
A	25	50	0
B	30	60	1
C	28	58	1
D	22	45	0
E	35	65	1

(Test point)

P: Age = 27, Salary = 55K

$$d(P, A) = \sqrt{(27-25)^2 + (55-50)^2} = \sqrt{4+25} = \sqrt{29} \approx 5.39$$

$$d(P, B) = \sqrt{(27-30)^2 + (55-60)^2} \approx 5.83$$

$$d(P, C) = \sqrt{(27-28)^2 + (55-58)^2} \approx 3.16$$

$$d(P, D) = \sqrt{(27-22)^2 + (55-45)^2} \approx 11.18$$

$$d(P, E) = \sqrt{(27-35)^2 + (55-65)^2} \approx 12.81$$

K=3 Nearest Neighbors

Top 3 nearest:

C (3.16) - Yes

A (5.39) - No

B (5.83) - Yes

Labels: Yes, No, Yes $\Rightarrow$ majority = Yes

Equation = Parametric
Else:
non-parametric

∴ Person P is predicted to have a wife (yes)

classification = majority vote of the neighbors
Regression = average (or weighted average) of neighbor values.

[Gradient Descent]

Ex-1: Perform one epoch and find the value (updated value) of m , $\alpha = 0.01$

USE SSR as loss function.

<u>height</u>	<u>weight</u>
2	3
4	5
6	7
8	9

Sol: $y = mx$
 $m = 0.1$

Squared Residual for T_1 :

$$(3 - m \times 2)^2 = (3 - 2m)^2$$

Squared Residual for T_2 :

$$(5 - 4m)^2$$

$$T_3: (7 - 6m)^2$$

$$T_4: (9 - 8m)^2$$

l = sum of squared residuals (SSR)

$$= T_1 + T_2 + T_3 + T_4$$

$$= (3 - 2m)^2 + (5 - 4m)^2 + (7 - 6m)^2 + (9 - 8m)^2$$

$$\frac{dl}{dm} = \frac{d}{dm} [(3-2m)^2 + (5-4m)^2 + (7-6m)^2 + (9-8m)^2]$$

$$= \frac{d}{dm} (3-2m)^2 + \frac{d}{dm} (5-4m)^2 + \frac{d}{dm} (7-6m)^2 + \frac{d}{dm} (9-8m)^2$$

$$\begin{array}{c} 0.1 \\ \downarrow \\ = 2(3-2m) \cdot (-2) + 2(5-4m) \cdot (-4) \\ + 2(7-6m) \cdot (-6) + 2(9-8m) \cdot (-8) \end{array}$$

$$= -233.6$$

$$m = m - \frac{dl}{dm} \times \alpha$$

$$= 0.1 - (-233.6 \times 0.001)$$

$$= 2.436$$

(Ans)

$$y = mx$$

$$= 2.436 \times 2 \approx 4.872$$

$$y = mx$$

$$= 0.1 \times 2 \approx 0.2$$

$$\uparrow$$

Actual was 3; so loss = $\left| \frac{4.872}{3} \right|$

$$= 1.624$$

$$\hookrightarrow \text{loss} = |(3 - 0.2)|$$

$$= 2.8$$

$$\uparrow$$

less loss